

Identification and characterization of a bacterial hyaluronidase and its production in recombinant form

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(Received 4 April 2016, revised 12 June 2016, accepted 13 June 2016, available online 4 July 2016)

doi:10.1002/1873-3468.12258

Edited by Stuart Ferguson

[Correction Note: This article was first published online on 4 July 2016 and was corrected on 19 July 2016. In the original publication the captions of Figures 2B and 2E contained some inaccuracies and have subsequently been corrected.]

Hyaluronidases (Hyls) are broadly used in medical applications to facilitate the dispersion and/or absorption of fluids or medications. This study reports the isolation, cloning, and industrial-scale recombinant production, purification and full characterization, including X-ray structure determination at 1.45 Å, of an extracellular Hyal from the nonpathogenic bacterium *Streptomyces koganeiensis*. The recombinant *S. koganeiensis* Hyal (rHyal_Sk) has a novel bacterial catalytic domain with high enzymatic activity, compared with commercially available Hyls, and is more thermostable and presents higher proteolytic resistance, with activity over a broad pH range. Moreover, rHyal_Sk exhibits remarkable substrate specificity for hyaluronic acid (HA) and poses no risk of animal cross-infection.

Keywords: hyaluronic acid or hyaluronan; hyaluronidase; *Streptomyces koganeiensis*

Hyaluronidases (Hyls) are a family of hydrolytic enzymes that degrade hyaluronic acid (hyaluronan, HA). In nature, Hyls are widely utilized as a delivery

instrument to facilitate the dispersion of venoms, toxins, bacteria and spermatozoa through interstitial spaces [1–3]. In humans, Hyls play essential roles in

Abbreviations

CSK, crystallization screening kit; EMBL-EBI, the European Bioinformatics Institute; ESRF, European synchrotron radiation facility; FDA, Food and Drug Administration; GCB, Granada crystallization box; HA, hyaluronic acid; Hyal, hyaluronidase; MAD, multiwavelength anomalous dispersion; MICINN, the Ministerio de Ciencia e Innovación; PBS, phosphate-buffered saline; PEG, poly(ethylene glycol); rHuPH20, recombinant human hyaluronidase; rHyal_Sk, recombinant *S. koganeiensis* hyaluronidase; SEC, size exclusion chromatography; Se-Met, seleno-methionine.

the daily turnover of HA; the products of HA are essential in maintaining and enabling a number of physiological processes such as cell division, cell connection, germ cell activity, DNA transfection, embryonic development, tissue repair and cell proliferation [4,5]. HA levels are maintained under very tight control, and elevated HA levels are directly correlated with various types of diseases. Studies of many cancer types, including pancreas, breast, colon and prostate cancer, have reported an abnormal accumulation of HA that generates a protective network surrounding and supporting cancer development [6–8].

Hyal has been used in clinical practice for more than 50 years to increase tissue permeability and promote the spread or dispersion of other coinjecting drugs, particularly local anaesthetics [9]; for use in the treatment of oedema, local inflammatory states, haemorrhoids, and chilblains; and as an adjunct to induce a significant reduction in the size of myocardial infarction [10–17]. These commercial, clinical formulations employ animal-derived Hyals (bovine or ovine testicular Hyal) [18,19] whose hydrolysis mechanisms have been studied extensively [20,21]. A bovine-derived Hyal (Amphadase) and ovine-derived Hyal injection (Vitrane) were first approved by the Food and Drug Administration (FDA) in 2004, followed by approval of a second bovine-derived Hyal (Hyalase) in 2005 [22]. However, although animal-derived Hyals are regularly used, their production is generally reported to be limited due to low purity, variable potency and uncertain safety. Such preparations are frequently contaminated with proteases, immunoglobulin and factors that increase capillary permeability and can induce IgE-mediated allergic reactions upon repeated administration, generally precluding their repeated use [23,24]. Notably, many case reports of adverse events have been published in the literature regarding the use of animal-derived Hyals with subsequence blocks [22].

In addition to animal-derived sources, a recombinant human Hyal called recombinant human hyaluronidase (rHuPH20) derived from the human sperm protein PH-20 and produced by mammalian cells was approved by the FDA in 2005 (Hyalenex) [22]. Although rHuPH20 is a highly purified and highly potent Hyal with a preferable safety profile, recombinant bacterial Hyals from nonpathogenic microorganisms have also attracted much interest within the medical industry. Bacterial Hyals, such as Amano HY-1 obtained from *Streptomyces hyalurolyticus*, are commercially available, but rHuPH20 and animal-derived Hyals remain the only FDA-approved Hyals [22].

The production of microbial Hyals is not adequate for industrial requirements; the highest yield, reported

by Mahesh, is 591 U·mL^{−1} [25,26]. Consequently, the development of a novel recombinant bacterial Hyal merits consideration to improve the safety and quality of Hyals.

This study reports the identification and isolation of a Hyal with high enzymatic activity from the non-pathogenic microbial strain *Streptomyces koganeiensis*. This enzyme was produced by a recombinant pathway in a nonpathogenic microorganism with outstanding safety levels [27]. The recombinant *S. koganeiensis* Hyal (rHyal_Sk) was fully characterized. X-ray structure determination revealed an unprecedented bacterial catalytic domain with advantageous properties compared with commercially available Hyals.

Material and methods

Autologous Hyal was obtained from *S. koganeiensis* and characterized as described in Data S1 – Section 1. *S. koganeiensis* Hyal gene prediction, cloning and expression as rHyal_Sk, as well as industrial-scale production, purification and full characterization, were performed as described in Data S1 – Sections 2–3.

Crystallization and structure determination

The 3D structure was resolved after analysing crystals of rHyal_Sk containing a single selenomethionyl residue by multiwavelength anomalous dispersion (MAD). Briefly, for the crystallization, the rHyal_Sk buffer (PBS) was exchanged for 25 mM Hepes/NaOH, pH 7.0 and 150 mM NaCl, in which the protein could be concentrated to 1.3 mg mL^{−1}. Initial crystallization screening was performed by the counter-diffusion method [28] in capillaries with a 0.1 mm inner diameter and a 50 mm length (390 nL) using prefilled GCB crystallization kits (Triana S&T, Granada, Spain; GCB-CSK, with 24 conditions in capillaries) as well as ammonium sulphate (KIT-AS-49) and a polyethylene glycol mixture (KIT-PEG448-49), both in the range of pH 4.0–9.0. A total of 72 experiments were carried out (36 assays at 20 °C and 36 assays at 4 °C). A typical pattern of crystallization obtained using the counter-diffusion technique is shown in Fig. S1A. Crystals appeared in several conditions, with all of them containing ammonium sulphate as the precipitant. Using KIT-AS-49 (3.0 M ammonium sulphate), crystals were obtained at pH 6.0, 7.0 and 8.0 at 4 °C and at pH 7.0 and 8.0 at 20 °C, showing that both temperature and pH play a role in protein solubility. Hits were also observed in conditions C5 (1.7 M ammonium sulphate, 3.5% poly(ethylene glycol)400 and 0.1 M Hepes/NaOH (pH 7.5), C7 (2 M ammonium sulphate and 0.1 M Tris/HCl (pH 8.5) and C9 (30% v/v poly(ethylene glycol)8K and 0.2 M ammonium sulphate) after

2 weeks. Crystals were soaked *in situ* with the mother liquor, containing 20% (v/v) glycerol, for 24 h and then flash cooled in liquid nitrogen. Data collection was carried out at beam lines ID23 and ID29 of the European Synchrotron Radiation Facility (ESRF, Grenoble, France). The best crystals grown in condition C9 diffracted X-rays to a maximum resolution of 1.5 Å (Fig. S1B). All attempts to find a molecular replacement solution using different models (such as PDB ID 2C3F, 2PE4, and 2WHT) and predicted models failed.

Next, we produced the Se-Met rHyal_Sk derivative following the already described protocol for expression and purification. Protein at 1.3 or 2.6 mg mL⁻¹ was subject to crystallization screening using the counter-diffusion kits, but with capillaries of 0.2 × 50 mm (1.6 µL). Hits were obtained under similar conditions as for the native protein but also in C3 (20% v/v poly(ethylene glycol)8K, 0.2 M magnesium acetate and 0.1 M sodium cacodylate) and C8 (26% v/v poly(ethylene glycol)4K, 0.32 M lithium sulphate and 0.1 M Tris/HCl (pH 8.5)). Because the crystals were still of small size (less than 50 µm in their longer dimension), the protein concentration was increased to 6.0 mg·mL⁻¹ to improve the final size. Rhombohedral crystals of approximately 100 µm in length were obtained in condition C9 and were used for MAD data collection at the Xaloc beam line of the ALBA spanish synchrotron radiation source (Granada, Spain) on a Pilatus 6M detector (Fig. S1C). Data were indexed and integrated using XDS and reduced with SCALA (39) of the CCP4 program suite [29]. The structure was determined using the MAD method routine, as implemented in the AUTO-RICKSHAW package [30]. In short, SHELXD [31] was able to find the two selenium sites located in the ASU, corresponding to the single methionine present in each monomer. The correct handedness for the substructure was determined with SHELXE [32] and the initial phases were calculated using MLPHARE [29], and improved by density modification with DM [33]. The partial model built automatically by ARP/WARP [34] located 255 amino acids (58% of the total unit cell content). Manual model building using COOT [35] and refinement using PHENIX [36] were employed to extend and improve the model, including several cycles of refinement using TLS parameterization [37]. Model quality was assessed with MOLPROBITY [38]. Statistics of data collection and refinement and properties of the final models are summarized in Table S1. All structure representations were generated with PYMOL [39].

Data deposition

rHyal_Sk sequence data generated in this study were submitted to GenBank under the accession number: KP313606. Protein 3D structure was deposited in EMBL-EBI, Protein Data Bank (<http://www.ebi.ac.uk/pdbe/>) under accession number: PDB ID 4UFQ.

Results and Discussion

Autologous Hyal from *S. koganeiensis*

An initial screen of nonpathogenic bacterial strains for Hyal activity [40] revealed the high extracellular Hyal activity of *S. koganeiensis* (Data S1 – Sections 1.1–1.2), confirming a previous study from 1978 in which the extracellular Hyal activity of *S. koganeiensis* was described [41]. Autologous extracellular Hyal was purified from *S. koganeiensis* (Hyal_Sk) culture supernatants (Data S1 – Section 1.3) and exhibited a molecular weight of 21.6 kDa, a pI of approximately 5.0 and an enzymatic activity of approximately 40 000 units·mg⁻¹ (Data S1 – Section 1.4).

S. koganeiensis Hyal gene prediction

Protease digestion of Hyal_Sk produced five peptide fragments, which were analysed by MALDI-ToF MS and Edman sequenced. A homology study of the five identified peptide fragments of Hyal_Sk against the GenBank database produced a match with a predicted protein (protein_id = ZP_06911952.1) of *Streptomyces pristinaespiralis* ATCC 25486. On the basis of the hypothetical protein (protein_id = ZP_06911952.1), a pair of degenerate primers was designed and used to amplify cDNA of the mRNA of *S. koganeiensis* (Data S1 – Section 2.1). Following several rounds of primer refinement, the final set produced a DNA encoding a predicted amino acid sequence matching that identified by Edman sequencing of the autologous Hyal_Sk (Data S1 – Sections 2.1–2.2).

Cloning and expression of rHyal_Sk

Cloning, protein expression and final purification of rHyal_Sk were performed and optimized in BL21 *Escherichia coli*. This process is detailed in Data S1 – Sections 2.3–2.6. The rHyal_Sk protein exhibited the same specific activity as the autologous Hyal_Sk (Data S1 – Section 2.7), but with higher protein yields, based on fed-batch fermentation of *E. coli* clones [BL21 (DE3)-pHyal_sk_SL] in 20 L reactors. The recombinant protein was found in the periplasm of *E. coli* BL21. This location is determined by the native N-terminal sequence which corresponded to the aminoacidic sequence amino acid sequence AGENGATTTFDGPVA. In the case of the autologous protein produced by *S. koganeiensis*, it displays at its N terminus a signal sequence (signal peptide) of 30 amino acids (amino acid position 1–30) (Data S1 – Section 2.2 and Appendix S3) which is cleaved

before being released in the supernatant, presenting at this point the following N-terminal part: AGEN-GATTTFDGPVA. This N-terminal sequence, which is thus found in the mature form of the protein in the *S. koganeiensis* culture supernatant, is not a typical sec-dependent recognition sequence. The above mentioned N-terminal sequences is not part of either the members of the post-translational (secB) or cotranslational (SRP or TAT) pathways (including mal, gIII, ompA, pelB, phoA, ompC, ompT, dsbA, torT, sufI, torA, STII, EOX, lamb, MglB, SfmC, TolB and MmAp). Furthermore, our N-terminal sequence is not a peptidase signal. However, as the rHyal_Sk is isolated with high yields from the periplasmic soluble portion, this sequence could be a sec-independent periplasmic protein translocation pathway in *E. coli*. The gold standard method for extracting specific periplasmic portions, osmotic shocks was used (Data S1 – Section 2.7). The rHyal_Sk in the periplasmic soluble portion was thus produced at a final concentration of approximately 2 g·L⁻¹ of culture medium with very high functional activity (more than 40 000 units·mg⁻¹), 670- to 750-

fold higher than the autologous Hyal produced by fermentation. Moreover, rHyal_Sk exhibited a substantially improved purity and safety profile compared with autologous Hyal_Sk, thus meeting the parameters required for pharmaceutical use (Data S1 – Section 3). Table 1 summarizes the main features of the novel rHyal_Sk.

rHyal_Sk characterization

Comparative functional tests were performed using rHyal_Sk and two commercially available Hyals approved by the FDA: Hyal BRP from bovine testis (EDQM, FIP Hyaluronidase, H1115000) and rHuPH20, a recombinant human enzyme (Hylenex) (Data S1 – Sections 3.14–3.19). A functional screening based on the measurement of HA cleavage using size exclusion chromatography (SEC) was first conducted (Data S1 – Section 3.16).

SEC analysis of HA cleavage by the three enzymes at different time points demonstrated that the catalytic efficiency of rHyal_Sk against HA was fully comparable to those of the commercial Hyals (Fig. 1).

Table 1. Summary of the main features of the new recombinant *S. koganeiensis* hyaluronidase (rHyal_Sk).

Protein original source	Bacterial
Technology	Recombinant
Gene expression host cell	Safe Bacterial cells
Expressed protein localization	Periplasmic space
Property	Soluble
Polypeptide chains	217 amino acids
Molecular weight (mass spectrometry) theoretical Mw: 21679.96	21679.98 ± 0.82 Daltons
Formula	C ₉₃₃ H ₁₅₀₈ N ₂₇₆ O ₃₁₆ S ₁
Protein concentration (mg·mL ⁻¹) by Lowry/bicinchoninic acid method	Pure form: ~ 0.8 mg·mL ⁻¹
Purity (SDS/PAGE) first dimension	Pure form: > 99%
Purity (SDS/PAGE) second dimension	96%
Storage	4 °C; –20 °C
Post-translational characteristics	Nonglycosylated
Tertiary structure	Cysteine disulphide bond free
Secondary structure (CD)	Same structure
Peptide mapping (by LC-MS)	92% (> 90%)
Identity by reference amino acid sequence	
Isoelectric point (theoretical pI: 5.5)	pI 5.2 ± 0.4
Purity by IEF	> 86%
N-terminal sequencing	AGENGA
Purity by SEC-HPLC	> 99%
Purity by RP-HPLC	> 99%
Proprietary HCP assays for <i>E. coli</i> (limit < 100 ppm)	6.5–8.8 ppm
DNA threshold system (limit ≤ 300 pg·mg ⁻¹)	19 pg·mg ⁻¹
Chromo-LAL	< 0.5 U·mg ⁻¹
Extinction coefficient	12 383 L × mol ⁻¹ × cm ⁻¹ ABS 280 (=1 g·L ⁻¹) 0.570
Capillary (Zone) electrophoresis purity	> 99%
Molecular aggregate (UV 250–350 nm analysis)	No aggregate
pH solution (in PBS 1×)	~ 7.9
Conducibility (in PBS 1× ~ 15 mS·cm ⁻¹)	~ 7 mS·cm ⁻¹
Turbidimetric hyaluronidase activity assay of Dorfman	~ 437 249.1 IU·mL ⁻¹

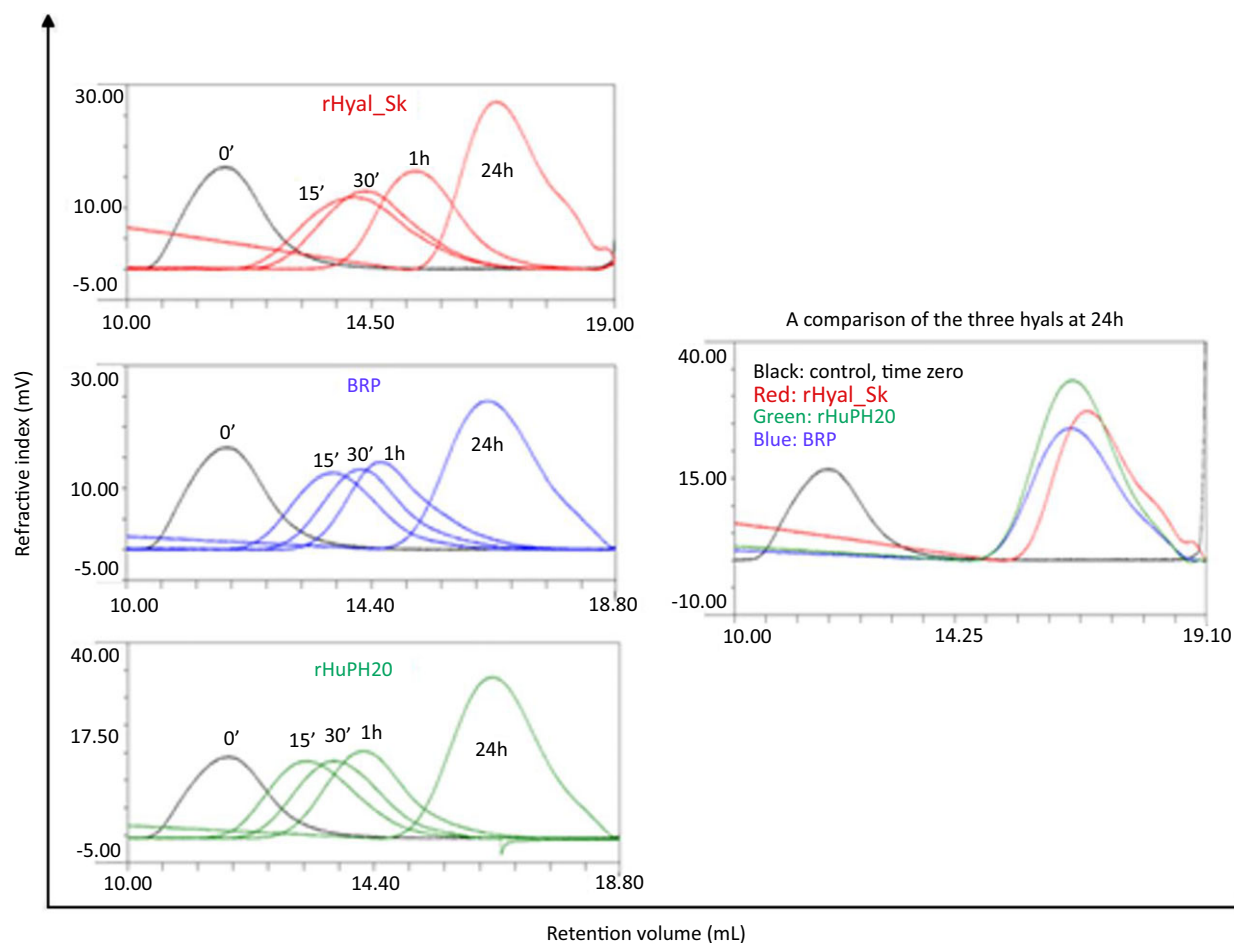


Fig. 1. Comparative functional assays of rHyal_Sk vs. BRP-Hyal and rHuPH20. Screening based on the measurement of HA cleavage using size exclusion chromatography (SEC).

rHyal_Sk, BRP-Hyal and rHuPH20 were treated with two different doses of Pronase E (a mixture of proteases isolated from the extracellular fluid of *Streptomyces griseus*), and the activities of the Hyals were measured at different time points. The enzymatic activity of rHyal_Sk was not altered at any analysed point, whereas BRP-Hyal and rHuPH20 exhibited reductions of activity of up to 40% compared with the initial time (Fig. 2A).

The stability of the two nonhuman Hyals, rHyal_Sk and BRP-Hyal, was evaluated in the presence of human or animal serum, which is known to effect the enzymatic activity of Hyals [42]. BRP-Hyal activity decreased by up to 40% after only 1 h of incubation with human or animal serum, whereas rHyal_Sk activity was unaffected (Fig. 2B).

These results suggest that rHyal_Sk may be effective for systemic applications, particularly cardiovascular and cerebro-vascular therapies [43,44]. A comparison of other physicochemical properties demonstrated that rHyal_Sk exhibited higher thermal stability and a

broader pH functionality range than either rHuPH20 or BRP-Hyal (Fig. 2C,D).

Finally, substrate specificity was also analysed. In contrast to BRP-Hyal, which hydrolyses both HA and other mucopolysaccharides (chondroitin sulphate, heparan sulphate), rHyal_Sk exhibited higher specificity and selectivity towards HA. The specificities of rHyal_Sk and rHuPH20 were similar (Fig. 2E and Data S1 – Section 3.17). Moreover, rHyal_Sk exhibits remarkable thermostability and activity over a broad pH range (Data S1 – Sections 3.18–3.19).

3D structure of rHyal_Sk

The 3D structure was resolved at 1.45 Å resolution after analysing crystals of rHyal_Sk containing a single selenomethionyl residue by MAD. This approach was adopted after the failure of all attempts to solve the structure by molecular replacement. Data were collected at the ALBA spanish synchrotron radiation

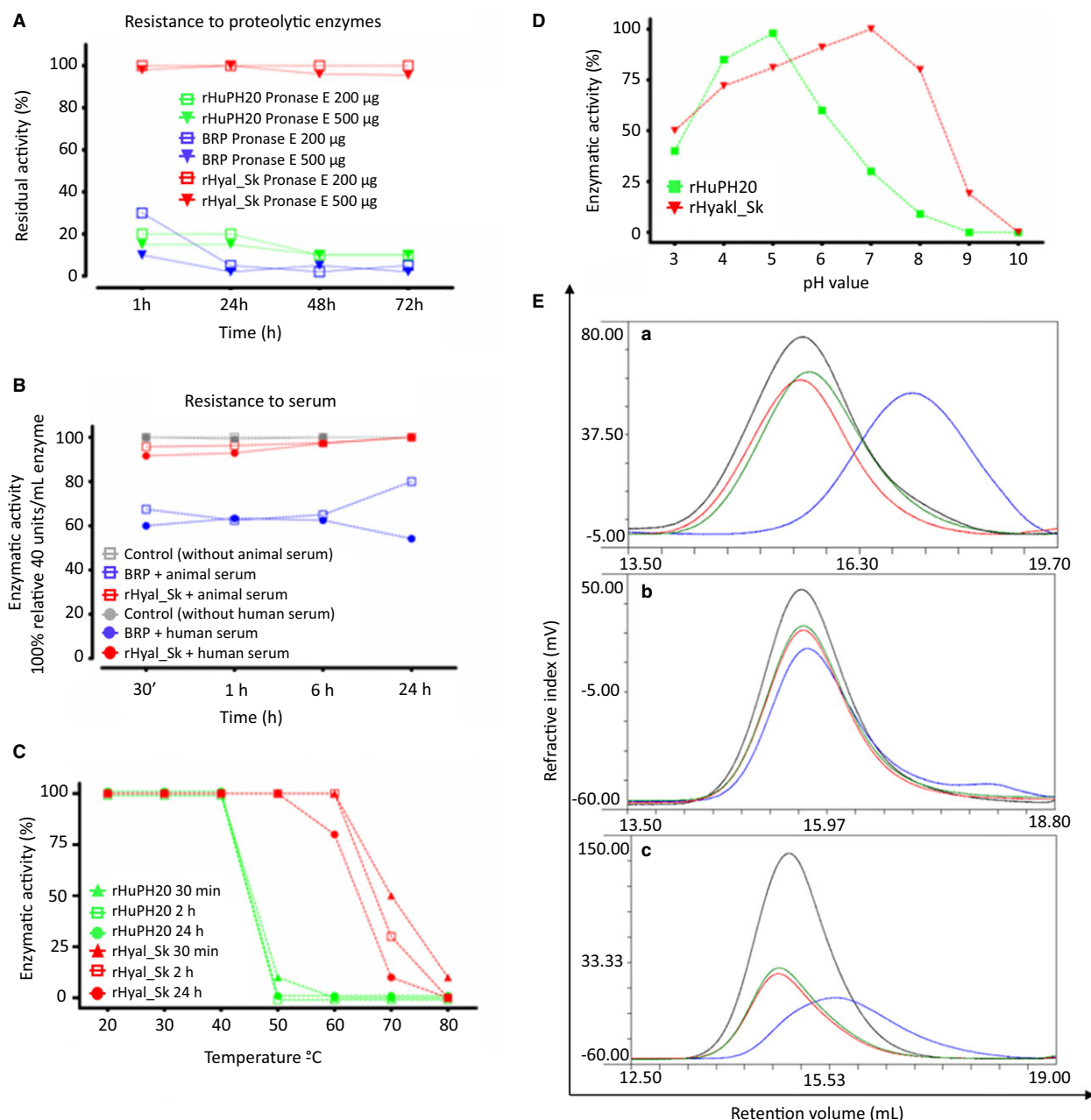


Fig. 2. Comparative functional assays of rHyal_Sk vs. BRP-Hyal and rHuPH20. Stability studies of rHyal_Sk vs. BRP-Hyal and rHuPH20 incubated with (A) Pronase E or (B) human and animal serum. (C) Evaluation of the enzyme activity at different temperatures. (D) Evaluation of the enzyme activity at different pH values. (E) Evaluation of the substrate specificity using (a) chondroitin sulphate A, (b) dermatan sulphate from porcine intestinal mucosa and (c) dermatan sulphate from shark cartilage (in black: control, red: rHyal_Sk, green: rHuPH20 and blue: BRP).

source (Spain). Two chains, A (residues 6–217) and B (residues 7–217), were located in the asymmetric unit with good definition of all amino acids. The 3D structure of rHyal_Sk includes a triple-stranded β -helix formed by three intertwined polypeptide chains around the threefold crystallographic axis (Fig. 3). Each chain forms 121 hydrogen bonds covering a total surface of approximately 12 379 Å². This fold has previously

been observed in other bacteriophages and phages encoding Hyals belonging to the PL16 Hyal family [45]. A unique feature of rHyal_Sk is the absence of the α -helix and capping domains observed in HyIP1 [46], HyIP2 [47] and the truncated forms of HyIP2 and HyIP3 [48] from *Streptococcus pyogenes*; rHyal_Sk exhibits only the central β -helix containing the active site for catalysis of HA degradation.

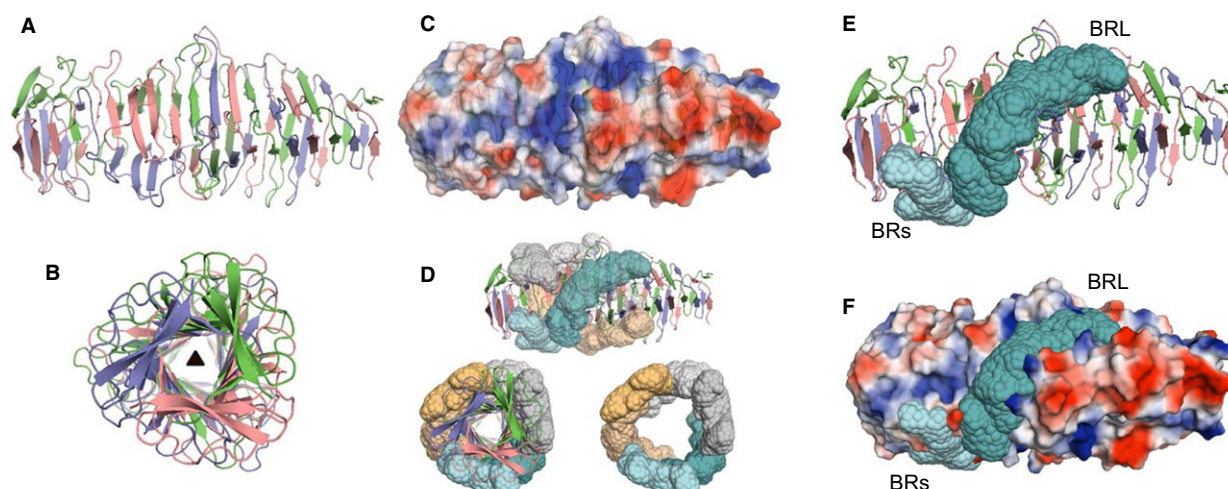


Fig. 3. Ribbon representation of the triple-stranded β -helix arrangement of rHyal_Sk. (A) Three intertwined polypeptide chains around the (B) threefold crystallographic axis. (C) Concentration of positively charged residues (blue) at the concave groove along the face of the triangular tube, where the HA chain may sit. In (D), the calculated cleft volume is represented as spheres in perspectives identical to those in (A) and (B). (E, F) Ribbon representation of the triple-stranded β -helix arrangement of rHyal_Sk, showing the putative binding regions, with one along the intertwined β -sheet (BRL) and a smaller one sitting on top of that one (BRs), represented as spheres. (B) shows the electrostatic surface representation of the trimer (the positively and negatively charged residues are coloured in blue and red respectively).

The right-handed β -helix is formed by β -strands grouped into four sheets: two consisting of intertwined strands of each monomer and the other two formed by each single chain in such a way that an almost continuous concave surface is created, interrupted only by several loops. The groove is rich in the positively charged amino acids arginine and lysine as well as asparagine, generating an electrostatic surface that complements the charge distribution of polysaccharides such as HA (Fig. 3C). The ligand-binding region is clearly observed when the cleft volumes are calculated. Two consecutive binding regions, one along the intertwined β -sheet (BRL) topped by the other, smaller (BRs approximately 27 Å), can be identified (Fig. 3D) [28,30,36–38,49–59].

Although both DLS and SEC data (Data S1 – Section 2.7 and Fig. S7) pointed to the monomeric species to be stable in solution, the X-ray structure shows that the active biological oligomerization state is an homotrimer. The observed deviation in DLS and SEC data can be explained due to the elongated-rod shape of rHyal_Sk, while standard calculations for DLS and SEC are based on globular shape proteins.

Conclusions

In summary, a novel extracellular Hyal of *S. koganeiensis* (Hyal_Sk) has been described for the first time. After analysing supernatants of a number of nonpathogenic bacterial strains, *S. koganeiensis* was selected because its supernatant exhibited high Hyal activity. Through a

clustering proteomic approach, the nucleotide sequence encoding the Hyal_Sk polypeptide was determined, which revealed that the encoded protein did not exhibit homology with known proteins. Upon cloning of the coding sequence, a novel soluble rHyal_Sk was expressed in the periplasmic portion of nonpathogenic BL21 *E. coli* cells at an industrial scale with a purity, activity and stability profile favourable for achieving high yields at competitive costs. Compared with commercially available Hyals, rHyal_Sk exhibits superior thermal and proteolytic resistance, a broader pH range of functionality and higher substrate specificity. In addition, the use of rHyal_Sk does not pose any risk of animal cross-infection. Therefore, due to its high stability, rHyal_Sk can be used not only for the preparation of pharmaceutical formulations to facilitate the subcutaneous administration of active substances but also for clinical medical treatments (such as surgery, ophthalmology, internal medicine, thrombotic events and tumours). The 3D modelling study revealed a new Hyal catalytic domain typical of the *Streptomyces* family (Fig. 3E,F); rHyal_Sk is the first and only bacterial Hyal to be fully described and characterized.

Acknowledgements

This research was funded by the MICINN (Spain) projects BIO2010-16800 (JAG), ‘Factoría Española de Cristalización’ Consolider-Ingenio 2010 (JAG) and Feder Funds (JAG). JJDM thanks MICINN for a

Ramon y Cajal Fellowship. We are very grateful to the staff at beam-line XALOC of ALBA synchrotron radiation sources (Spain), ID23-1, -2 and ID29 of the ESRF synchrotron radiation sources (France) for support during data collection. We thank Prof Mark Bradley and Prof Paul Lizardi for their support and proofreading. These studies were approved and supported by Fidia Farmaceutici S.p.A. and NanoGetic S.L.

Author contributions

LM, SV, JAG and CS conceived and designed the experiments. LM, JAG, AP, FS, MP performed the experiments. LM, JAG, SaC, LuM, RDip, MP and SC analysed and interpreted the data. EP-M, MC-M, AP and FS contributed reagents/materials/analysis tools. EP-M, MC-M, AP and FS performed the computation work. LM, SP, JDUB, RMSM, JJD-M wrote the paper, prepared figures and/or tables. LM, JAG, SP and JJD-M reviewed drafts of the paper. CS approved the article for publication.

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Supporting information

Additional Supporting Information may be found online in the supporting information tab for this article: **Fig. S1.** Crystallization experiments of recombinant *S. koganeiensis* hyaluronidase (rHyal_Sk) in capillaries using 30% v/v poly(ethylene glycol)8K, 0.2 M ammonium sulphate as precipitant (A). Isolated crystals of rHyal_Sk and the seleno-methionine (**Se-Met**) derivative crystals are shown in (B) and (C) respectively.

Table S1. Statistics of data collection and refinement and properties of the final models.

Data S1. Supporting Materials and methods.

Fig. S2. Hyal units measure per milliliter of *S. koganeiensis* culture at different time points.

Fig. S3. SDS/PAGE (12%) protein pattern of the fractions obtained at the end of each purification step compared with supernatant protein pattern.

Fig. S4. Schematic illustration of inverse PCR (IPCR).

Fig. S5. Recombinant protein expressed by *E. coli*, in the different fermentation parameters, analysed by electrophoresis in SDS/PAGE (12%) after staining with Coomassie Brilliant Blue G-250 (Bio-Rad, 161-0406).

Fig. S6. Analysis of the protein content after each purification step by electrophoresis in SDS/PAGE (12%) after staining with Coomassie Brilliant Blue G-250 (Bio-Rad, 161-0406).

Fig. S7. Extract of the Viscotek-TDA chromatogram obtained for rHyal_Sk L.002/11.

Fig. S8. Analysis on agarose gels of HA depolarization.

Fig. S9. Circular dichroism measurements of autologous Hyal vs rHyal_Sk.

Fig. S10. Mass spectrometry analysis of rHyal_Sk after separation in Reverse phase (RP)-HPLC.

Fig. S11. The isoelectric point determined by comparison with the reference standards' isoelectric points (pH 3-10 IEF Marker, SERVA).

Fig. S12. Chromatogram of rHyal_Sk by HPLC.

Fig. S13. Chromatogram of rHyal_Sk by RP-HPLC.

Table S2. Summary of the substrate specificity and selectivity of rHyal_Sk vs BRP-Hyal (EDQM, FIP Hyaluronidase, H1115000) and rHuPH20 (recombinant human enzyme; Hylenex).

Table S3. Substrate specificities of rHyal_Sk, Hylenex and BRP against other polysaccharides.

Appendix S1. *In silico* amino acid sequence of the ~ 550 bp band amplified from *S. koganeiensis* genomic DNA.

Appendix S2. Nucleic acids: In lower case, the noncoding region located in the 5' and 3'; In italics, the coding region 5' and 3' identified with the latter processing steps; In bold, the START and STOP codons; Underlined, sequences corresponding to the oligonucleotide in 5' MesFor2 (in bold conservative nucleotide differences) and MesINTf / MesINTr; and Underlined and in bold, restriction sites for *KpnI*. Aminoacids: Highlighted in bold, regions corresponding to the results of previous protein sequencing.

Appendix S3. *In silico* amino acid sequence of sequenced hyal gene after amplification from *S. koganeiensis* genomic DNA.

Appendix S4. CLUSTAL 2.1 Multiple Sequence Alignments of Hyaluronoglucosaminidase protein (genomic locus code YP_006266806.1) of *Actinoplanes* sp.; Predicted protein ZP_06911952.1 (genomic locus code ZP_06911952.1) of *Streptomyces pristinaespiralis* ATCC 25486; STSU_30255 hypothetical protein (genomic locus code ZP_10072726.1) of *Streptomyces tsukubaensis* NRRL18488; and *In silico* amino acid sequence of sequenced hyal gene after amplification from *S. koganeiensis* genomic DNA.

Appendix S5. The map of the pHyal_sk_SL plasmid: ori origin of replication of *E. coli*, *Ap* gene for ampicillin resistance, T7 promoter, *lacI* gene (*lacI* repressor), hyaluronidase (Hyal) gene encoding the *S. koganeiensis* Hyal, *NdeI* and *EcoRI* unique restriction sites used for cloning the hyal.

Appendix S6. The nucleotide sequence of the ORF encoding the hyal protein of *S. koganeiensis* cloned into vector pET21b(+) (Novagen) and expressed in *Escherichia coli*.